



Delhi Public School Megacity, Kolkata

Class 4 Session 2026-27

Subject: Science

Topic: Friction as a Force

Please note that all assignments will be completed in the presence of the subject teacher during class hours. Students are not allowed to copy these assignments into their exercise books in advance. In case of absence, students may refer to the assignments to ensure that their work is up to date.

Q1) Why does a ball rolling on the ground stop after sometime?

Ans: A ball rolling on the ground stops after sometime due to friction. It is the force produced when one surface (ball) tries to slide/roll over another surface (ground). It always opposes motion and tries to bring the moving object to a halt.

Q2) Write any three uses of friction in our daily life.

Ans: Three uses of friction in our daily life are:

- i) It prevents us from slipping on the floor while walking.
- ii) Friction between the lead of a pencil and paper helps us to write.
- iii) It helps us to hold things.

Q3) How can you say that friction produces heat?

Ans: When we rub our hands in winter, they become warm. This is due to the fact that our hands being rough when slide over each other generates friction which in turn generates heat.

Q4) On a rainy day, people walk carefully on the road. Why should we walk carefully on a wet road?

Ans: We should walk carefully on a wet road because water reduces friction between our shoes and the ground, thus making the surface slippery.

Q5) Give reasons:

a) We apply oil to moving parts of machines.

Ans: Oil fills in the grooves of the surface of machines making its surface smooth thereby reducing friction and causing less wear and tear.

b) We use anti-skid tiles in bathrooms.

Ans: We use anti-skid tiles in bathrooms to increase friction and prevent people slipping on wet floors.

Q6) You are trying to slide a heavy cardboard box filled with books across a carpeted floor, but it is very difficult to move.

1. Why is it hard to move the box on the carpet?

It is hard to move the box on the carpet because the rough surface of the carpet creates high friction with the bottom of the box, making it difficult to slide.

2. Mention one way to make the movement of the box easier.

We can place smooth materials underneath the box, such as cardboard or plastic sheets, or wheels to reduce friction.

3. If you switch to a smooth wooden floor, would the box be easier or difficult to move? Why?

It would be easier to move. A smooth floor has less friction (fewer rough spots interlocking), requiring less force to push the box.